Measurement 1: Atomic Force Microscopy

Measurement Exercise for Physicist-Engineer Students

Abstract. Since its development in 1986, Atomic Force Microscopy (AFM) has become a leading tool of sample characterization in academia as well as in the industry. Surfaces can be imaged with nanometer resolution, and the chance to harness extra interactions enables wide-range electrical characterization, from work-function mapping to dopant density determination. This student measurement is an introduction to AFM measurements which will be conducted via the Semilab AFM-1000 device in the Semilab HQ Dexagon Showroom.

Warning. The AFM machine has a high-voltage power supply for the piezo actuators and a laser module for cantilever detection. Working with the machines is safe as long as you work cautiously and follow the safety instructions. However, careless work can lead to accidents such as losing your eyesight. It is therefore mandatory to read and understand the safety rules and guidelines before entering the laboratory. Furthermore, repeat reading the warnings before each action in the lab and strictly follow the instructions described in the *Measurement Tasks* section. Always ensure the safety of yourself and others during measurements. If you are unsure about anything, ask the lab supervisors for help and always follow their warnings and instructions.

Theoretical summary

I

1 Introduction

Let's imagine that we are provided with a silicon cantilever with a pointy tip at one end like in Figure I.1.1. If we are to gently approach a surface with the pointy part perpendicular to the sample surface, first the cantilever would remain horizontal. Getting closer would cause some attractive forces to act between the tip and the sample, and the cantilever would bend down towards the sample. If we were to keep getting closer, the surface would slowly start to repel the cantilever. At first, it would become horizontal, and then it would start bending upwards, away from the sample.

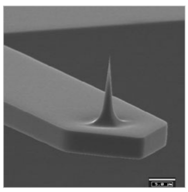

Figure I.1.1. SEM image of AFM cantilever tip. Image source: Nano and More.

The tip-sample interaction causing the behavior described above can be modeled by the Lennard-Jones potential shown in Figure I.1.2. This essentially tells us that close to the sample there is an attractive and even closer there is a repulsive regime.

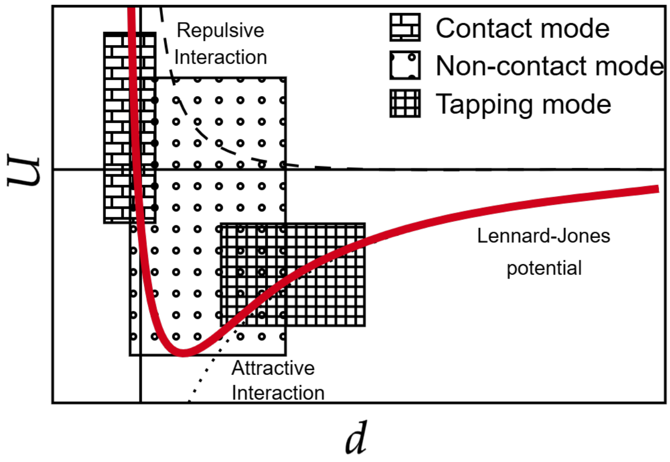

Figure I.1.2. Lennard-Jones potential, the model of cantilever-surface interaction potential in atomic force microscopy. Horizontal scale is the tip-sample distance. Image source: Semantic Scholar.

It turns out that using a piezoelectric crystal, such gentle movements can be performed in a controlled fashion. But how do we determine the amount of deflection? If we shine a laser on the backside of the cantilever and observe the reflected beam from far away, cantilever movements on the order of nanometers can be easily sensed. These techniques enable us to investigate surfaces on the nanoscale, and the machine bringing all these ideas together is referred to as an Atomic Force Microscope (AFM). The main components of an AFM are depicted in Figure I.1.3.

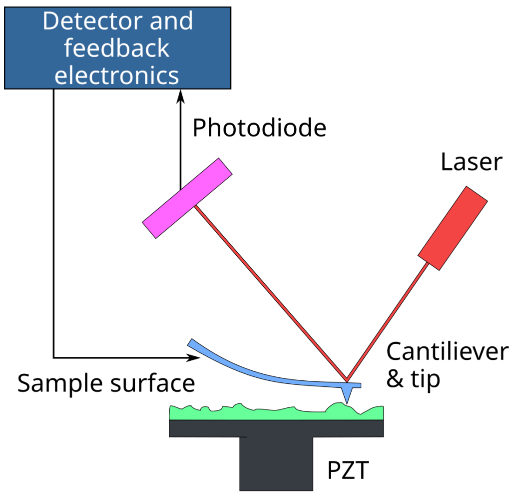

Figure I.1.3. Main parts of a general purpose AFM system. Cantilever for surface topography measurement, laser and photodiode for cantilever positioning and piezo actuators for sample and/or cantilever positioning. Image source: Wikipedia.

Surface scanning means that using two extra piezo crystals, the cantilever can also be moved in the plane perpendicular to the sample surface. Storing the bending signal would provide us with some information about the sample surface, but the periodic stress would destroy the sharp cantilever tip quickly. To overcome this, we have to turn our attention towards basic regulation theory.

2 DC mode

The simplest mode of operation, often referred to as *DC mode* means pushing the tip into the sample and dragging it over the scanning area. But instead of measuring the bending at a constant z-position the amount of bending is kept constant and the vertical position is continuously adjusted, resulting in a constant interaction force between the tip and the sample surface (see Figure I.2.1.).

The tip is moved laterally and the bending is measured. Once the tip gets close to a hump, the bending signal gets larger (the interaction force increases) and the tip must be elevated. The opposite case is when it runs above a trench, the bending signal decreases (since the interaction force decreases). The tip must be pushed closer to the sample to keep the interaction strength constant.

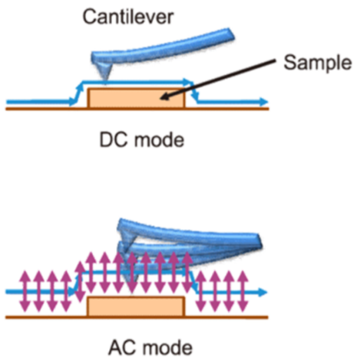

Figure I.2.1. Two modes of operation: DC mode means the tip is in direct contact with the surface for the whole measurement, whereas in AC mode the surface is only periodically contacted thus reducing the interaction time and the resulting damages.

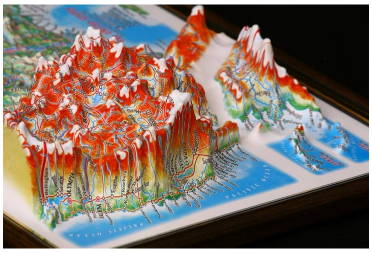

Figure I.2.2. Raised relief map. The height of the given geological feature is visible from both height and colour data.

3 PI Controller

This positional fine-tuning can be realized via a *proportional-integral (PI) controller*. Let the $b(t)$ optical detector signal (bending signal) be the time-dependent *reference signal*, as the scanning in the xy-plane is in progress. If the desired amount of bending b_0 - the so-called *set-point* - is known, then the *error signal* $e(t)$ is calculated as

$$e(t) = b(t) - b_0.$$

The goal of the PI controller is to ensure the constant bending; thus it adjusts the z-position according to the equation

$$z(t) = P \cdot e(t) + I \cdot \int_0^t e(t')\ dt'.$$

The accuracy and speed of the control are highly dependent on the parameter settings.

By performing a raster scan with constant interaction forces and continuous height adjustment, these Δz adjustments can be measured at each $\Delta z(x_i, y_j)$ point, and eventually a "raised relief map" (Figure I.2.2.) is outlined, which resembles the *surface topography*.

Harnessing the PI regulation capabilities, the DC mode becomes a suitable method to scan stiff surfaces. However, tip degradation will still be an issue, and softer samples - e.g. biological cells - can be easily torn apart if the interaction forces are too big.

4 Frequency Sweep

What if we were to use the piezo crystal to force the cantilever to oscillate? The optical detection scheme would still work, but the laser spot would also oscillate. Using the lock-in measurement technique, we can measure the frequency component of the optical signal that corresponds to the cantilever excitation. The schematic amplitude and phase response of a cantilever for a wide range frequency sweep can be seen in the upper part of Figure I.4.1. To achieve maximal oscillation amplitude, the cantilever must be excited at its resonant frequency.

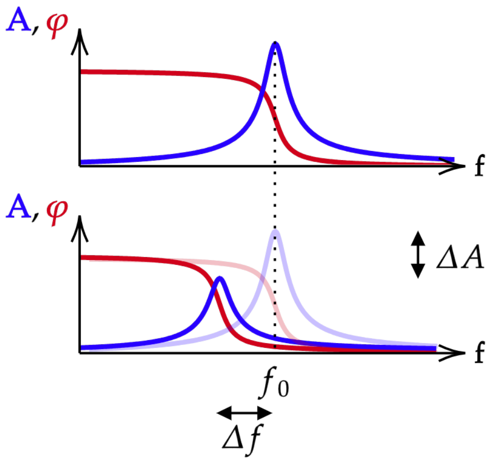

Figure I.4.1. Frequency and phase shift and amplitude damping experienced by the cantilever in the attractive regime.

The resulting data sets can fitted via the functions

$$A(f; A_{\text{fit}}, f_0, Q, A_0) = A_{\text{fit}} \frac{f}{\sqrt{f^2 + \left(\frac{Q}{f_0}\right)^2 \cdot \left(f^2 - f_0^2\right)^2}} + A_0$$

and

$$\varphi(f; f_0, Q, \varphi_0) = \arctan\left(Q \frac{1 - \left(\frac{f}{f_0}\right)^2}{\frac{f}{f_0}}\right) + \varphi_0.$$

This way the f_0 *resonance frequency* and A *resonance amplitude* are both determined, alongside the Q *quality factor* of the oscillation plus the constant shifts A_0 and φ_0. The latter may be used to shift the phase to zero at the resonance.

These amplitude and phase curves represent the situation when there is negligible interaction between the sample and tip because these are separated at a macroscopic distance. What happens if we start approaching the surface?

5 AC mode

The main experimental observation is that the resonance curves shift to lower frequencies once the forces become significant, as illustrated in the lower part of Figure I.4.1. This height interval is the *attractive regime*, which is the focal point of AC imaging. It is important to note that further approaching the tip will lead to a shift towards higher frequencies. This happens in the *repulsive regime*.

If the resonance curve shifts to the left (lower frequencies), then the forced oscillation at the original f_0 resonance frequency will have a smaller amplitude

$$\Delta A = A_{\text{osc.}}(z) - A_{\text{fit}} < 0.$$

The amplitude decrease measured at the resonance frequency is the useful signal in AC AFM imaging.

> **Remark.** Another effect contributing to the decrease is the presence of non-conservative forces which damp the oscillation. In this approach, the effect of the

shift and the damping cannot be separated; however, the net amplitude decrease can be used to define at which height we expect to reach the surface.

In this case, the error signal reads as $e(t) = A_{\text{osc.}}(t) - A_0$ and yet again the continuous adjustment of the vertical position enables topography mapping! Compared to the DC mode, now surface contact has been minimized, and next to the amplitude the phase can also be monitored, which often provides extra insight into the material composition of the sample. Another key difference is that in the DC mode we were adjusting the interaction force between the cantilever and the sample, but in the AC mode we determine the oscillation amplitude. The *amplitude set-point* is usually defined as a percentage. Setting 30 % amplitude set-point means, that the tip is set to the height z above the (x_i, y_j) point for which the measured amplitude is 70% of the amplitude measured far from the surface. Or equivalently:

$$\overline{A_{\text{osc}}(t)} = 0.7 \cdot A_{\text{fit}}.$$

In summary the amplitude set-point is the reference, *the amount of decrease in the oscillation amplitude which characterizes the surface.*

During this measurement exercise, the AM-AFM measurement is realized, which is often referred to as *tapping mode*, since the tip-sample contact is reduced; it only occurs within a small portion of the period.

6 PI Tuning

To thoroughly understand the effect of the parameter settings, let us briefly revise what happens in an over (small P, large I)- or under-damped (large P, small I) feedback circuit. At the beginning of the raster scan, the cantilever is oscillating on the surface, at the amplitude set-point, thus the PI controller's output indicates no change in the z-position. Once a high enough surface element is met by the cantilever, the oscillation gets damped. The error signal is said to be proportional to the ΔA changes in amplitude: its proportional (P) component suddenly increases, but it is immediately suppressed by the small value of P. Plus, due to the large timescale of integration, the error signal is smoothed over in the integral component (I). Once the integral part accumulates enough large error inputs, the cantilever gets elevated and the amplitude increases back to the set-point value. For trench-like structures the opposite happens: above the trench, there is no damping due to the increased tip-sample distance, thus the oscillation amplitude gets larger. Similarly, the P component is not dominant enough and the integral part delays the response. These slow feedback settings may even make the cantilever get into continuous contact with the surface. As a result, the topography image gets smeared, and the contrast around the edges of sample structures becomes significantly lower.

The solution seems straightforward: let's increase the proportional part. This is reasonable up to a certain extent. Once the feedback becomes too fast, even the simple electrical noise in the error signal is enough to induce unwanted high-frequency z-modulation. This means that besides being excited by the shaker piezo, the cantilever is dragged up and down by the z-piezo as the feedback circuit struggles with minimizing the error signal and the resulting image becomes noisy. In this case, the P parameter must be decreased up to the point where no large fluctuation is present. The schematic output for too fast, too slow, and ideally set PI circuits when the tip encounters a positive and a negative unit step can be seen in Figure I.6.1.

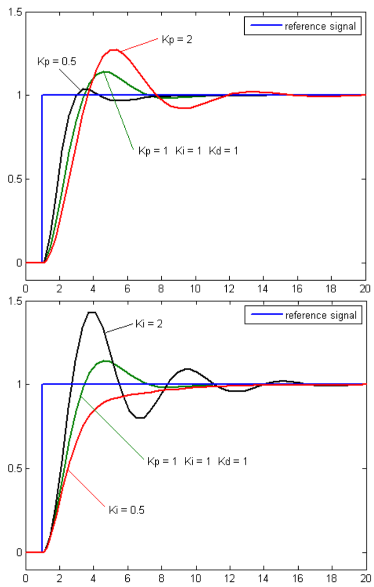

Figure I.6.1. Unit-step response of a PID controller upon changing P and I respectively while keeping the remaining parameters unchanged.

7 Scan Correction

AFM is all fun until it is only the basic theory, but in practice, the piezo crystals exhibit many undesired behaviors (drift, creep, hysteresis), the cantilever oscillation spectrum is full of side peaks, the optical detector is noisy, and the sample is never perpendicular to the tip. Finding all the ideal settings in an AFM device for a given sample and tip system is more like an art. However, some of the system problems distort the measured images in a very typical fashion. Figure I.7.1 provides a great outline of the most common imaging artifacts simulated for a rectangular reference sample. Studying these will help you to find and eliminate the typical causes of faulty imaging.

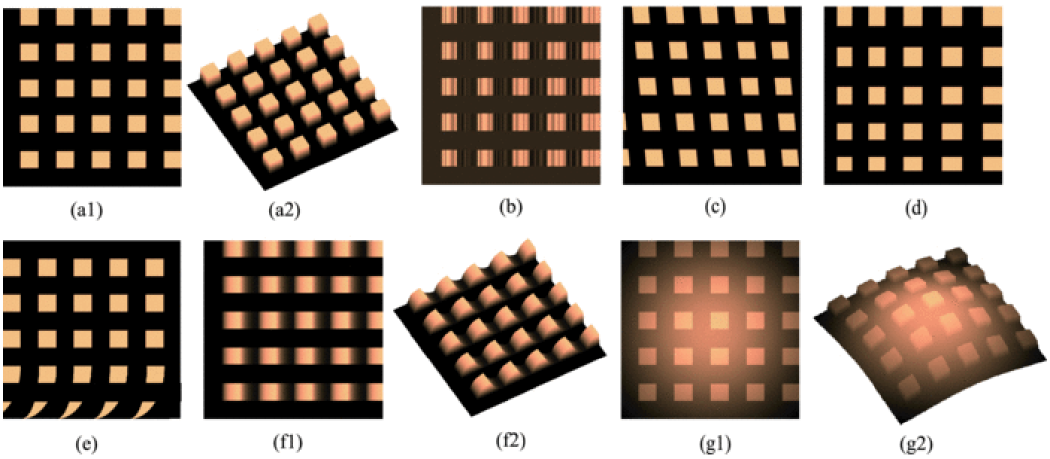

Figure I.7.1. Simulated AFM image artifacts. (a1 and a2) Ideal grating without artifacts. (b) Artifacts caused by vibrations, (c) x-to-y cross-coupling motions, (d) the presence of hysteresis at the lateral axes, (e) the creep and drift effect, (f1 and f2) insufficient vertical feedback controller bandwidth, and (g1 and g2) xy-to-z cross-coupling motions.

Once the device and the scan settings have been configured properly we will receive the scan data as a matrix. A common experience is, that barely anything is visible, just like in I.7.2.a). This is the result of sample tilt, the tip and the sample are not orthogonal. Thanks to the active feedback (and a large enough z-piezo range) the measurement can be performed, but the resulting topography will also be tilted. The simplest solution is to find the best fitting plane and subtract it from the measurement result, like in I.7.2.b). For many cases this is sufficient, but there is a chance that after the tilt correction the lines of the scan will not be properly aligned.

Then the strategy is changed, and line correction has to be performed on the data. We iterate through the lines of the scan, and fit each individual line with a polynomial of order p. The order is usually changed between 1 and 6 iteratively in a trial-and-error process. The lowest possible order, where the misalignment disappears, is desirable. For the example measurement, the effect of first and second order line correction can also be observed in Figure I.7.2.c) and Figure I.7.2.d). In this example, $p = 2$ yields excellent results without the danger of overfitting the lines and corrupting data validity.

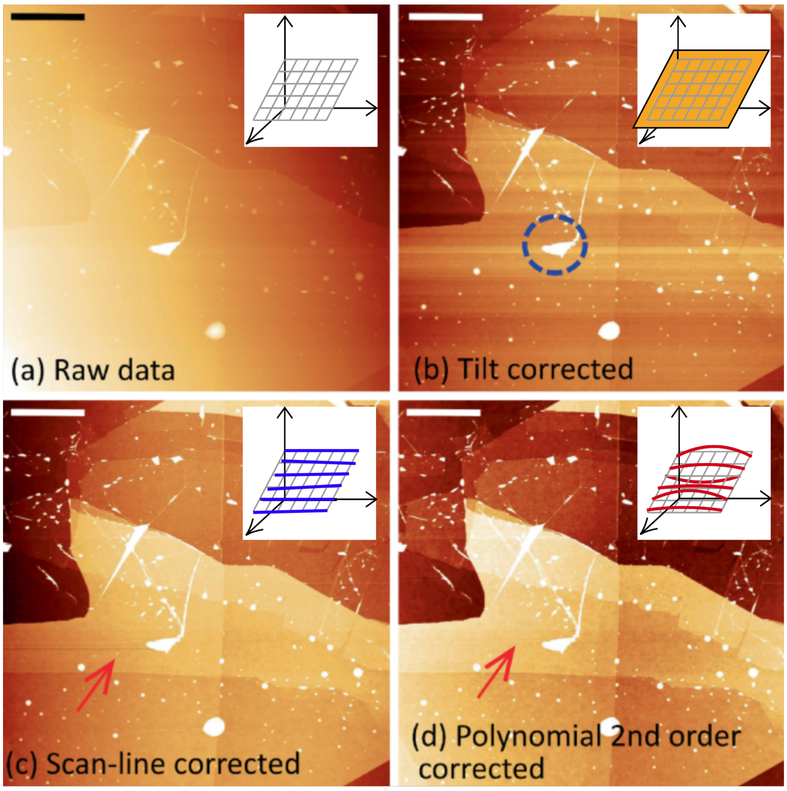

Figure I.7.2. Fundamental scan correction methods. a) Raw data. b) Tilt corrected data. c) Data after first order line correction. d) Data after second order line correction. For each subplot schematics graphics were added which help visualize the correction. Raw data sampled on rectangular grid, and best fitting plane, lines and parabolas. Image source: Springer Nature.

8 Summary

In conclusion, a successful AFM measurement consists of careful

- Mode selection (DC or AC).
- Tip analysis.
- Set-point selection.
- PI setting optimization.
- Data correction.
- Data interpretation.

In the next chapter, you will be introduced to the device calibration, which helps set up a mathematical connection between the piezo voltages and tip displacement, and then a short review is provided of the role of AFM in the semiconductor industry.

Measurement methods

II

1 Calibration

In order to estimate how much the cantilever tip moves when the positioning piezo actuators experience a given voltage signal, calibration has to be performed. Based on the orientation of the actuator, we distinguish between *lateral calibration* for the xy plane and *vertical calibration* for the z axis.

For this very purpose, *calibration sample* are available, which exhibit a periodic pattern with known *pitch size* and *step height* (Figure II.1.1.). Calibration is an iterative process. A sensitivity K for each direction is already stored in the measurement program, and the new value is always based on the current one. All we need to do is measure the sample, estimate the pitch size or step height, and multiply the stored sensitivity with the proportion of the nominal value and the measured value. In mathematical form, this reads as

$$K^{\star}_{x_i} = K_{x_i} \cdot \frac{P_{\text{nominal}}}{P_{\text{meas}}}.$$

For this exercise the μ-Masch TGXYZ01 calibration grating will be used. This is a nested periodic topography created by SiO_2 features on a Si wafer. As it can be seen in Figure II.1.2., there are two outer halves and four inner quadrants. The step heights for all patterns are 100 nm $\pm$ 2 %, whereas inner quadrants have a pitch size of 5 $\pm$ 0.1 μm, and the outer halves have a pitch size of 10 $\pm$ 0.1 μm.

The measurement software provides us with an automatic calibration module. After proper leveling of the data, the periodic features are recognized, and a grid is fitted on the feature centers. The fitting results in an accurate estimate of the lateral periods. Calculating the histogram of the corrected data points, two Gaussian peaks arise because our sample has two distinct levels. Fitting these with the well-known formula, the difference of the expected values provides us again with an accurate estimate of the step height.

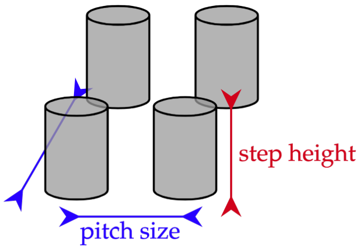

Figure II.1.1. Schematic drawing of cylindrical calibration features with pitch size lateral period and step height feature height.

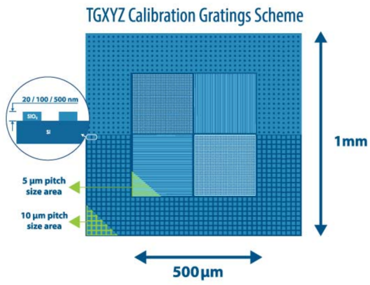

Figure II.1.2. Feature map of the calibration sample.

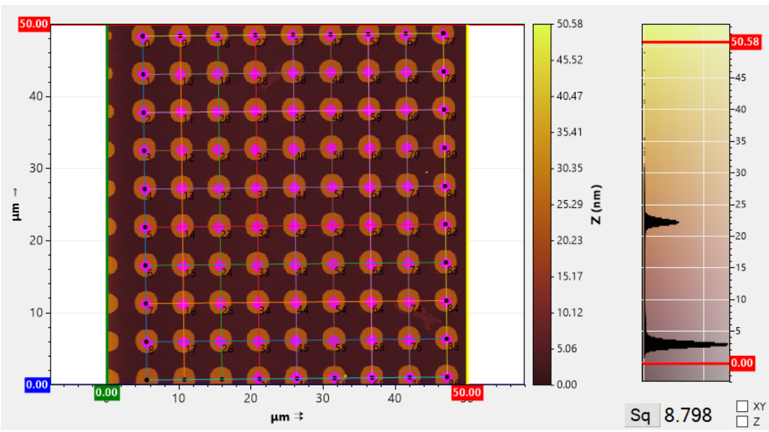

Figure II.1.3. Example output of the calibration module. Lateral calibration is performed via grid fitting (left-hand side) whereas vertical calibration is based on fitting the histogram (right-hand side).

2 Semiconductor Industry

In the semiconductor manufacturing process, AFM plays a key role with its high lateral resolution. A key application of an AFM machine is to perform process control, i.e. measure a sample throughout the many steps of semiconductor device fabrication and provide feedback on the product quality and uniformity of the process. The fundamental measures of a sample can be the geometrical properties of the sample patterns (critical dimension) and the inhomogeneity of the flat parts (roughness).

2.1 Critical Dimension

Form follows function - in many cases, the micro- and nanomanufactured samples exhibit a periodic structure with purpose, as in waveguides or diffraction gratings. This implies that the manufacturing process is only successful if the periodic features have the desired relation with the specific wavelength. To ensure this, an AFM scan can be performed, and after proper data correction, the linear dimensions can be measured for all individual structure elements. Calculating the descriptive statistics for period, width, and height means estimating the sample's critical dimensions.

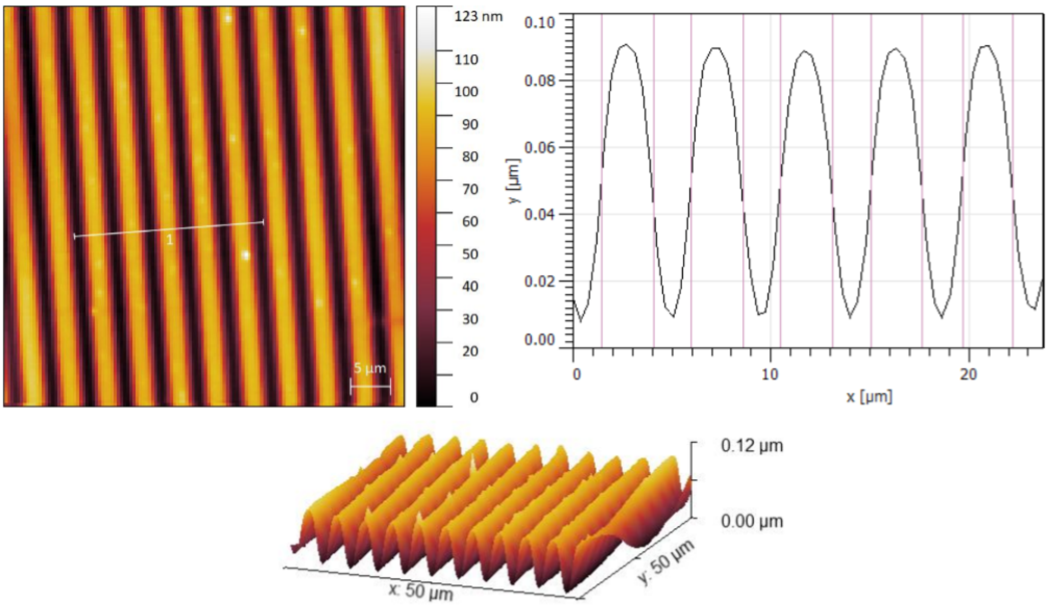

Figure II.2.1. AFM image of the platinum resistor. Using a higher order line correction the artifacts have disappeared and a line profile can be extracted. Using various methods the geometric feature properties are easily determined.

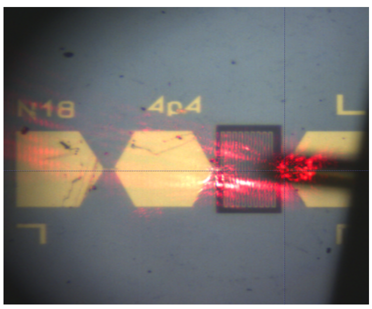

Figure II.2.2. Camera image of the initial form of an aspiring neural circuit element, with three electrodes, the platinum meander resistor, the AFM cantilever and some diffracted laser spots.

In Figure II.2.2. you can see an image of three electrodes and a meander-shaped platinum resistor. Together with a capacitor (added to the circuit later) the resistor is responsible for tuning the frequency and strength of oscillation in a neural circuit element. Using the Semilab AFM-1000 the center of the meander region has been scanned, and the analysis revealed an average central width of 2.60 [μm] and height of 80 [nm].

2.2 Roughness

The topography of a surface can be characterized by the spatial frequencies present. Traditionally, the components are divided into three groups, namely

1. Form: low-frequency components.

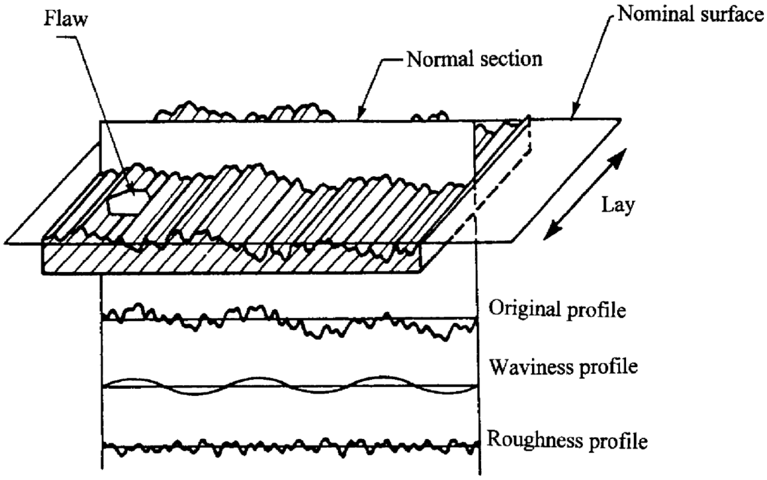

Figure II.2.3. Visual representation of waviness and roughness on a sample surface. Note how both quantities are defined with respect to the best fitting plane! Image source: Science Direct.

2. Waviness: medium-frequency components.

3. Roughness: high-frequency components.

Using an AFM the usual measurement scale is 50-100 [μm] for scan width and height. Thus for the nanoscale the terms waviness and roughness (microroughness) are widely applied. Figure II.2.3 shows how these can be interpreted; for a nano-scale sample, one could refer to the sample topography as waviness and the superimposed surface inhomogeneity as roughness. The latter has become a key notion in the semiconductor industry, since surface smoothness often has a high impact on the performance of the devices fabricated on the given wafer.

> **Remark.** A prime example of a structure where surface microroughness is a determining factor is the MOS transistor. Later in your studies in the framework of solid state physics, you will learn about the transistor, which is a three-terminal device used for switching and amplifying electrical signals. These devices have been around for about eighty years; they are the fundamental building block of small-scale electronics. Following Moore's law from the seventies, semiconductor manufacturers were working hard to fit more and more transistors onto a single chip by halving the size of a transistor every eighteen months. The size reduction has led to the point where the gate length is comparable to the depletion layer width of the source and drain junctions. This gives rise to multiple undesired effects, such as subthreshold leakage, which is the increase in the off-state current. One way to fight this was to increase the dopant density in the channel substrate. This, however, leads to smaller charge carrier mobility and larger electrical fields normal to the Si-SiO_2 interface. The latter leads to more frequent charge carrier collisions on the interface, which further reduces electron mobility. A detailed analysis based on scanning probe microscopy has shown that refining the proportion of the chemical agents used for the RCA cleaning step helps maintain low surface roughness. This diminishes scattering and leads to an increased mobility, ensuring that size reduction does only minimally interfere with device performance.

Multiple methods have been developed to characterize the roughness of a nano-scale surface. The most widespread indicators are shown in Table II.2.1, where z_i iterates over all the data points acquired during the scanning procedure. Naturally, these indicators are only evaluated after thorough correction.

Symbol	R_{a}	R_{q}	R_{sk}	R_{ku}
Definition	$\frac{1}{N}\sum_{i=1}^{N}\lvert z_i-\bar{z}\rvert$	$\sqrt{\frac{1}{N}\sum_{i=1}^{N}(z_i-\bar{z})^2}$	$\frac{\frac{1}{N}\sum_{i=1}^{N}(z_i-\bar{z})^3}{R_q^3}$	$\frac{\frac{1}{N}\sum_{i=1}^{N}(z_i-\bar{z})^4}{R_q^4}$

Table II.2.1. Standard measures of the surface roughness.

Remark. Observe the formulas in Table II.2.1. Based on their meaning and the subscript in the names, how would you call them? Have you encountered them earlier? What could you know for sure if they were evaluated for a Gaussian distribution?

Measurement tools

III

List of Measurement Tools.

- Semilab AFM-1000 machine.
- Computer.
- Pair of tweezers.
- Disposable gloves.
- TGXYZ-01 calibration sample.
- Patterned wafer fragments.
- Si wafer fragments.
- GaAs wafer segments.

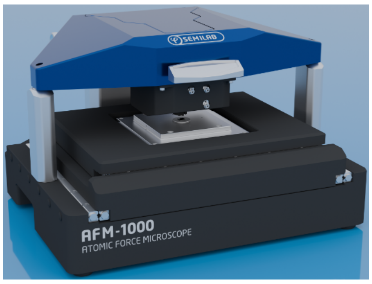

Figure III.2.4. Semilab AFM-1000 machine.

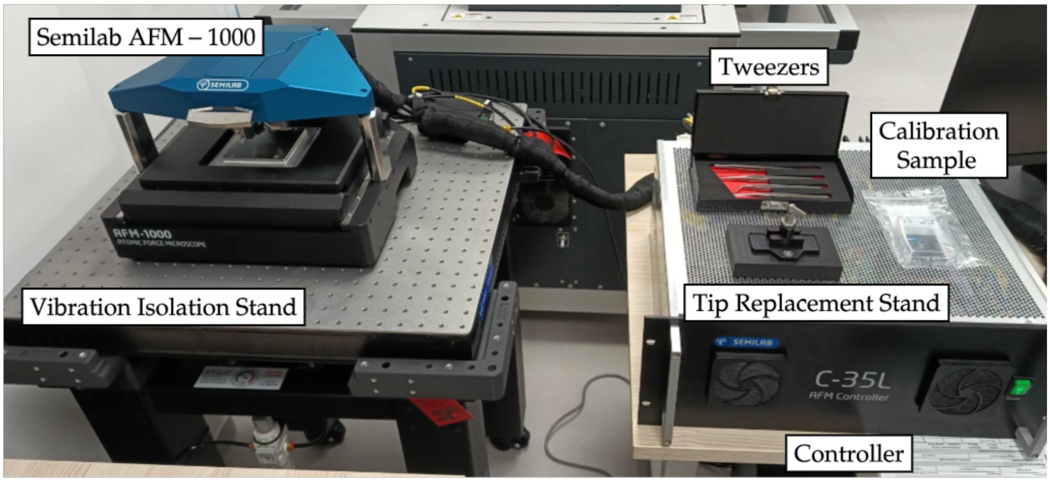

Figure III.2.5. AFM measurement setup at Semilab showroom.

Measurement Tasks

IV

Task 1. Prepare for the laboratory exercise.

1. Before the measurements, please create a folder of the format

$$< \text{Name}_1 > \_ < \text{OM ID}_1 > \_\_\_ < \text{Name}_2 > \_ < \text{OM ID}_2 >$$

at the path

C:\\SDF\\result.

You are expected to work in this folder throughout the whole laboratory exercise. When performing a measurement, the created path and desired filename can be set using the controls shown in Figure IV.1.1. The measurement results will be provided to you via a shared online folder, ensuring safe data transfer. It is strictly forbidden to browse through the results of others!

2. put on a pair of disposable gloves. Please refrain from touching your skin or any other source of contamination while wearing the gloves!

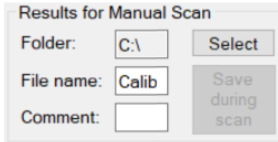

Figure IV.1.1. Excerpt of the Scan tab on the Operation window showing file saving settings.

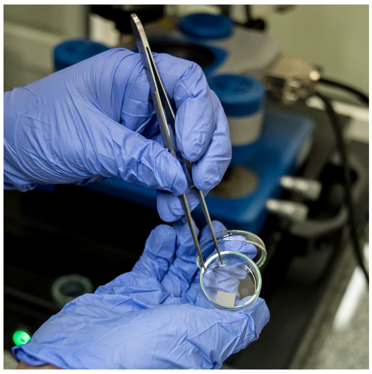

Figure IV.1.2. Always wear gloves when handling the microscope and the samples!

Warning. When the top of the AFM is open, the optical path of the readout laser is uncovered. Looking into the laser light directly or indirectly can cause severe damage to the eyes! Be careful when working with the system top open.

Warning. In the Semilab AFM-1000 machine, the sample positioning is performed by a motorized stage, which also moves the sample along the z-axis, whereas tip scanning is performed by lateral piezo actuators operating in the top part of the machine. Since the top part can move independently of the stage, stage elevation can put the sample at a dangerous height. Always close down the top part carefully, ideally using a fully retracted sample stage!

Task 2. Perform a frequency sweep with the machine top open and closed.

Open up the machine and make sure the top stays in place. Open the SDF software. Navigate to the Operation Window and the Frequency Sweep tab. Check if the controller is connected to the computer and make sure the light level is acceptable. Perform a frequency sweep by pressing the Full Sweep button (Figure IV.2.1). Observe and save the measurement results.

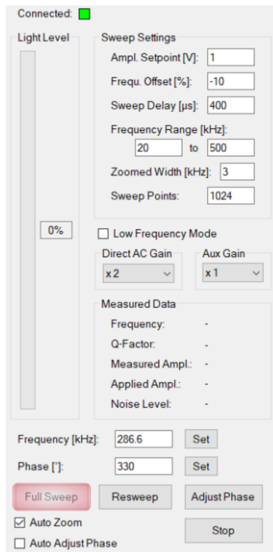

Figure IV.2.1. Excerpt of the Frequency Sweep tab on the Operation window.

Hint for the Lab Notes. Right-click on both the full and the zoomed-in spectrum to store the sweep results.

Close the machine top carefully and perform the sweeping procedure again. Do you see any difference?

Hint for the Lab Report. Think about the potential light sources generating a signal in the optical detector. What could have interfered with the sweeping procedure with the machine open?

Task 3. Calibrate the XY and Z piezo actuators.

Navigate to the Scan tab of the operation window. Make sure the stage is in a parked position. Open up the machine and make sure the top stays in place. Take the TGXYZ-01 calibration sample out of its box using a pair of plastic-tipped tweezers. Put it on the motorized sample stage of the machine, then gently close the top.

Warning. The sample stage of the machine is magnetic, and it attracts the metallic sample holders on which the calibration samples are mounted. Pay close attention when loading the samples, as they may easily slip out of the tweezers' grip!

Warning. ONCE THE SCANNER STATE CHANGES FROM PARKED POSITION IT IS STRICTLY FORBIDDEN TO TOUCH THE MICROSCOPE! Any mechanical excitation of the surroundings, the sample holder, or the machine housing can lead to a tip-sample crash event, which leads to the destruction of the tip, contamination of the sample, and may also cause damage to the AFM mechanical system.

Warning. When maneuvering to a sample position that lies outside the current scan range, the AFM tip has to be elevated to a safe distance above the sample. Make sure elevation is performed before moving the stage. When in doubt, contact the lab supervisor!

Using the motorized stage and the camera image, try to aim for a region of the calibration sample with circular features. Perform a tapping mode scan of the sample with four features visible along both axes following these steps and the figure IV.3.1:

1. Switch on the Update checkbox for Line Traces. This provides you with real-time feedback on the cantilever z-position and the error signal.

2. Set the Scan Parameters according to the table IV.3.1. This resolution provides us with enough detail to perform calibration but also maintain a reasonable measurement time.

Width [μm]	Offset [μm]	Points	Rate [Hz]
22.5	0	128	1

Table IV.3.1. Scan Parameters for Calibration.

3. Make sure AC mode is selected among the scanner settings. Setup a damping setpoint of 30%.

   How would you explain this setting in your own words?

4. Approach the sample with the tip using the respective Scanner Motor button. Wait until the Motor Position changes from Unknown to Surface.

5. Perform your fist AFM measurement by using the Scan button Scanner Motor Command. During the scan observe Line trace excerpts.

Is it what you would expect based on the camera image? Do you see any low frequency changes?

Figure IV.3.1. Excerpt of the Scan tab on the Operation window. Highlights show scan indicators, scan settings and regulation settings.

Once the scan has successfully finished, navigate to the Result window. From the file browser (Figure IV.3.2.) select the latest measurement. Right-click on the Trace Topography and select the Open for Analysis option. Press the correction button, then perform the following steps (Figure IV.3.3):

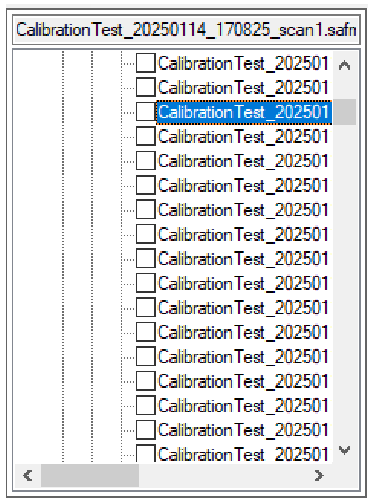

Figure IV.3.2. Excerpt of the File Browser widget on the Result tab.

1. Add a Line of order 1 on the Correction page.

   What happened to the image? Did the correction reveal any useful details? Do you see any trouble with the resulting image?

2. Add the LineCorr on the Analysis page.

   What happened to the image now? How did the histogram change during the first three steps? What do you think could be the difference between the two line fit operations?

3. Add the Segmentation on the Analysis page.

4. Add the XYCalibration on the Analysis page.

5. Add the ZCalibration on the Analysis page.

The result should be like in Figure II.1.3.

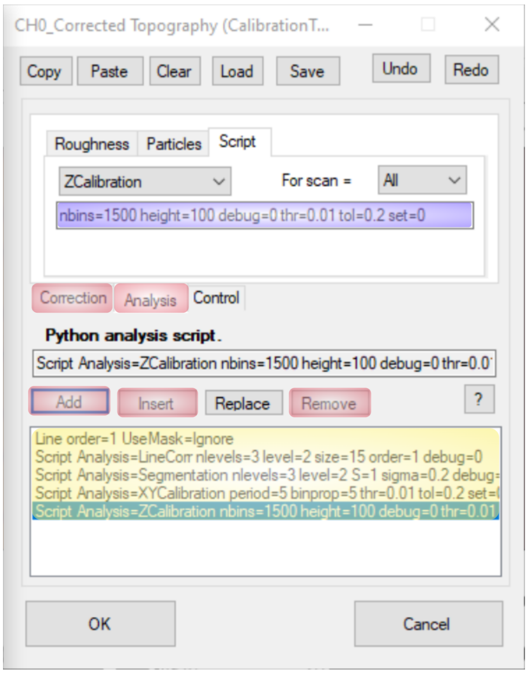

Figure IV.3.3. Excerpt of the Correction window accessible via the analysis window.

Hint for the Lab Report. In the lab report, please include the corrected measurement data, the old and new calibration parameters, and an explanation of the calibration procedure.

Task 4. Examine the effect of changing the PI parameters.

Start another scan on the sample with the initial PI settings. During the measurement, keep the I parameter fixed and change the P value. Make it much smaller and much bigger than the optimal value. Then set the P back to optimal and perform the same experiment with the I parameter.

Hint for the Lab Notes. Scan at least 8-10 lines with each setting. Also, record the error signal!

Hint for the Lab Report. In the lab report, please include line profiles from all six parameter pairs. Showcase not only the topography, but also the error signal. Explain in your own words what you see. For instance, how does the standard deviation of the error signal depend on the settings? Is there a parameter pair where the topography and the error signals "switch places"?

Park the sample stage. Once it is finished, gently open the microscope and unload the sample!

Task 5. Perform a critical dimension measurement.

Now that calibration has been performed, you are ready to accurately measure sample geometry. Pick a patterned wafer and load it into the microscope. Gently close the top. Using the camera, identify a region on the sample where you see a periodic pattern. Repeat the steps described for the calibration scan with the selected sample.

Hint for the Lab Notes. Show the scan with at least three features visible. PI settings need to be optimized for this task separately!

Hint for the Lab Report. Using your preferred evaluation approach, perform critical dimension measurement on the corrected data. Please provide a spreadsheet with the evaluated width, height, and period of the features. Also, calculate basic descriptive statistics, including *mean*, and *standard deviation*.

Task 6. Perform a roughness measurement.

Pick a silicon and a gallium-arsenide wafer fragment. Load one of them into the microscope. Gently close the top. Using the camera, identify a debris-free region. Using the scan parameters in Table IV.6.1. perform a measurement. Repeat the same procedure for the other sample.

Width [μm]	Offset [μm]	Points	Rate [Hz]
1	0	256	1

Table IV.6.1. Scan Parameters for Roughness Measurement.

Hint for the Lab Notes. Show both scans after tilt correction. Based on your initial analysis, do you believe what you see? If yes, what is the main difference between the roughness of the two samples?

Hint for the Lab Report. Compare your measurement results with literature. Which sample has smaller roughness? Think about the semiconductor industry and provide an explanation for the difference. Could you measure the expected values? Please explain any discrepancy between the expected and the measured data.

Supplement

V

For the lab report, there are two ways to evaluate the measured datasets. The first one is using Gwyddion, a free software developed for analyzing scanning probe microscopy data. The second one is using the gwyfile package in Python, which enables simple reading and editing features for properly formatted binary files. Since hands-on evaluation methods are available only to a limited extent, the latter method is only recommended to the very dedicated.

Gwyddion

Click here for tutorials, starting at downloading and installation and leading up to advanced data evaluation features.

Python

Code.

```
import gwyfile # If missing : pip install gwyfile
import matplotlib.pyplot as plt
import numpy as np

def read_safm_file(path, channel):
    # Load the gwyddion - formatted file
    gwy_container = gwyfile.load(path)
    # Parse the required fields
    meta = gwy_container[f"/{channel}/meta"]
    data = gwy_container[f"/{channel}/data"]
    # Convert data array to 2D numpy array
    data["data"] = np.array(data["data"])
    data["data"] =
       data["data"].reshape(data["xres"],data["yres"])
    return meta, data

if __name__ == "__main__":
    path = # File path here.

    meta, data = read_safm_file(path, 0)

    plt.figure(figsize=(4,4))
    plt.imshow(data["data"],
               origin="lower",
               cmap="afmhot",
               extent=(0,data["xreal"],0,data["yreal"]))
    plt.colorbar(label=r"$Z$ [nm]", shrink=0.675)
    plt.xlabel(r"$X$ [m]")
    plt.ylabel(r"$Y$ [m]")
```

```
    plt.tight_layout()
    plt.show()
```